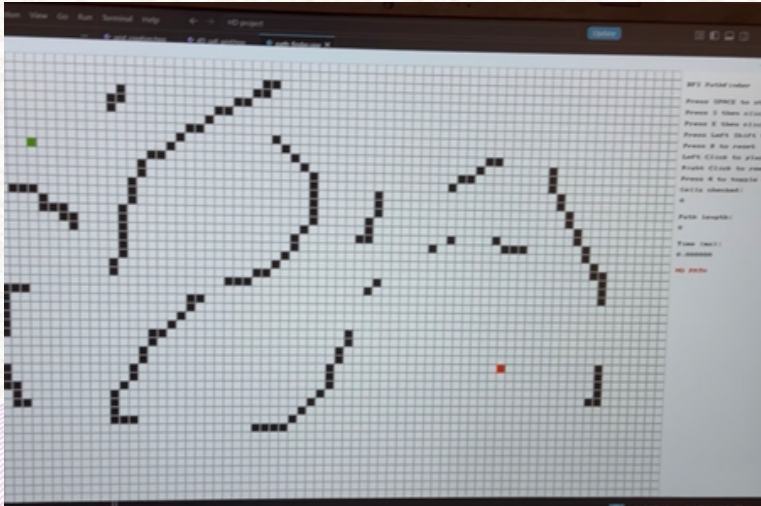


# PATHFINDER CODE/C++

Personal Project - Charlie Truscott

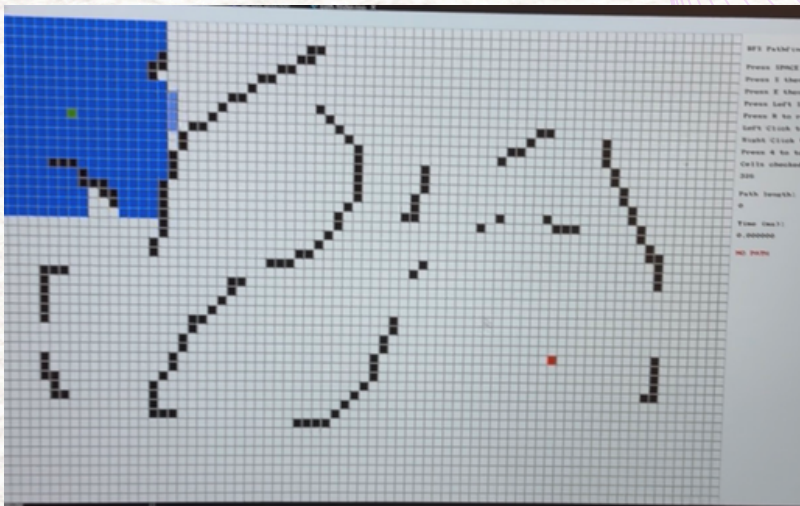
## First Creating the maze/path



## Project Overview

Built a grid-based pathfinding visualiser in C++ using the SplashKit library, implementing Breadth-First Search (BFS) to guarantee the shortest path across an  $80 \times 50$  cell grid. BFS was split into three functions: start, step, and trace, so the search could be animated frame-by-frame via an adjustable speed slider.

## Second running the algorithm



## Data Structure & Algorithm

The grid was modelled as a 2D array of Cell structs, each storing position, state, colour, and a parent reference for backtracking. An enum (EMPTY, WALL, START, END, VISITED, PATH) kept state management type-safe. BFS used a fixed-size array as a manual queue, exploring all equally-distant cells before advancing.

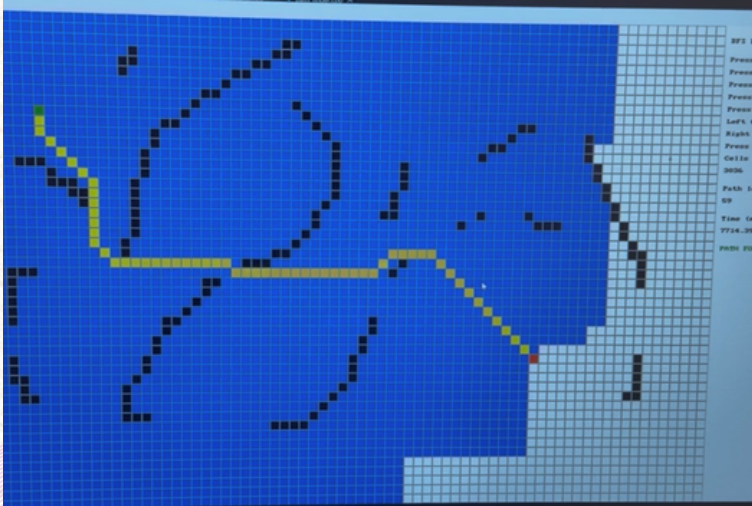
## Personal Insights

This was a HD (high distinction) project for my coding subject FIT1045. Being a beginner to C++/coding, I wanted to push myself with the basics I was taught into making a project I wanted to do.

# PATHFINDER CODE/C++

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Finally finding the shortest path



## Onscreen Control Board

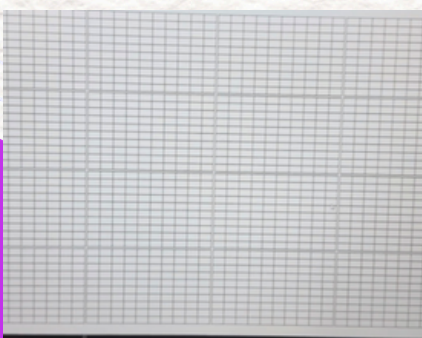
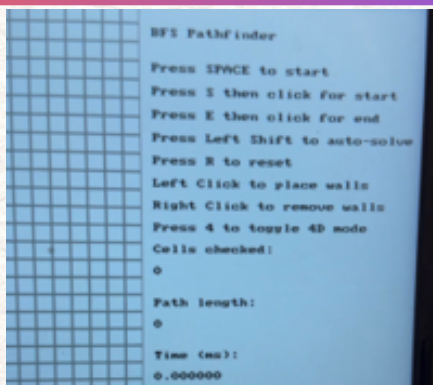
**SPACE** - auto-solve

**S / E + click** - set start / end

**Left Shift** - instant solve

**Left / Right click** - draw / erase walls

**R** - reset grid



## Diagonal Movement

Diagonal movement was added to extend the pathfinder beyond cardinal directions. This introduced corner clipping cutting through the gap between two diagonally adjacent walls resolved by checking the two shared cardinal neighbours before allowing the move.

## Mouse Interaction

Pixel coordinates were translated into grid indices by dividing by cell size. Wall drawing was improved by tracking mouse position across frames and linearly interpolating between the previous and current cell, so no gaps appeared when the cursor moved quickly.

## 4D Array Experiment

An experimental extension visualised a 4D array as a  $4 \times 4$  arrangement of  $10 \times 10$  sub-grids. The layout renders correctly, but extending BFS to pathfind across the extra dimensions is still unresolved a feature left unfinished rather than working.